



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Spatio-Temporal Analysis of Surface Urban Heat Islands Using High-Resolution Remote Sensing and SPDE-INLA: the Case Study of Bologna, Italy

Marika D'Agostini¹, Massimo Ventrucci¹, Andrea Ranzi²

¹ Department of Statistical Sciences "Paolo Fortunati", University of Bologna

² Regional Agency for Prevention, Environment and Energy of Emilia-Romagna, Italy

Motivation

- Analyse the differences and temporal changes in **Land Surface Temperature** (LST) and **Surface Urban Heat Islands** (SUHI) in the municipality of Bologna, Italy.
- Evaluate how **land cover** categories influence SUHI and LST

Data

- LST data**: 30 m resolution USGS **Landsat 8** (Collection 2, Level-2, Tier 1) Band 10 summer images (June-August, 2013-2024)
- SUHI**: defined pixel-wise as the difference between LST and the mean LST of non-urban pixels, identified using the ISPRA 2023 land use map
- Covariates:
 - Altitude** data from the DTM 5x5 (Geoportal of Emilia-Romagna Region)
 - Month** of remote sensing image acquisition
 - Land cover** (ISPRA 2012-2023), with vegetation types (coniferous, broadleaf, shrub, herbaceous) classified as urban vs. non-urban

Methodology

- A) Bayesian Spatially Varying Coefficient (SVC) Model for LST**

$$y_i(s) \sim \mathcal{N}(\mu_i(s), \tau^2)$$

$$\mu_i(s) = \mathbf{X}_i(s)\beta + u(s) + t_i \cdot v(s)$$

$$u(s) \sim GRF(0, Q_u^{-1}) \quad v(s) \sim GRF(0, Q_v^{-1})$$

$$\tau \sim \text{Gamma}(a, b) \quad \beta_k \sim \mathcal{N}(0, \tau_k^2) \quad k = 0, 1, \dots, 20;$$

- B) Bayesian SVC Model for SUHI**

$$z_i(s) \sim \mathcal{N}(\mu_i(s), \tau^2)$$

$$\mu_i(s) = \mathbf{X}_i(s)\beta + u(s) + t_i \cdot v(s)$$

$$u(s) \sim GRF(0, Q_u^{-1}) \quad v(s) \sim GRF(0, Q_v^{-1})$$

$$\tau \sim \text{Gamma}(a, b) \quad \beta_k \sim \mathcal{N}(0, \tau_k^2) \quad k = 0, 1, \dots, 20;$$

where

- $y_i(s)$ = LST at day i and pixel s
- $z_i(s)$ = SUHI at day i and pixel s
- $\mathbf{X}_i(s)\beta$ = linear predictor for day i and pixel s
- $u(s)$ = **spatially varying intercept** (residual spatial effect)
- $t_i \cdot v(s)$ = **spatially varying linear effect of time** (long-term summer linear trend)
- Spatially varying intercept** and **spatially varying linear effect of time** modelled as **latent Gaussian random fields** using the **SPDE** (Stochastic Partial Differential Equation) approach
- Posterior excursion sets at 95% probability** [1] for $t_i \cdot v(s) > 0$ and $t_i \cdot v(s) < 0$ to identify warming and cooling regions
- Inference performed with **INLA** [2]

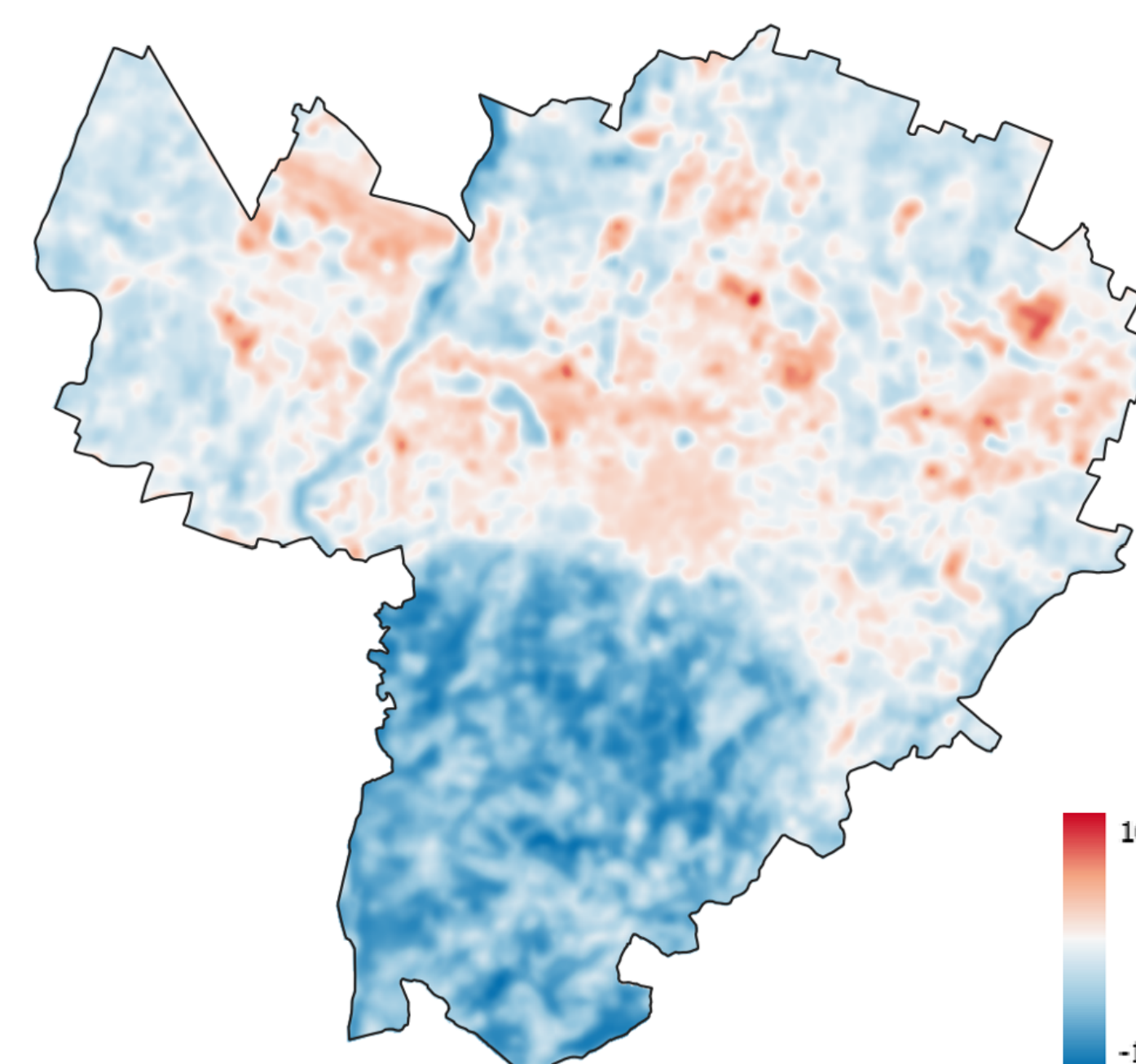
Web App

marika-dagostini.github.io/graspa25-suhi-bologna

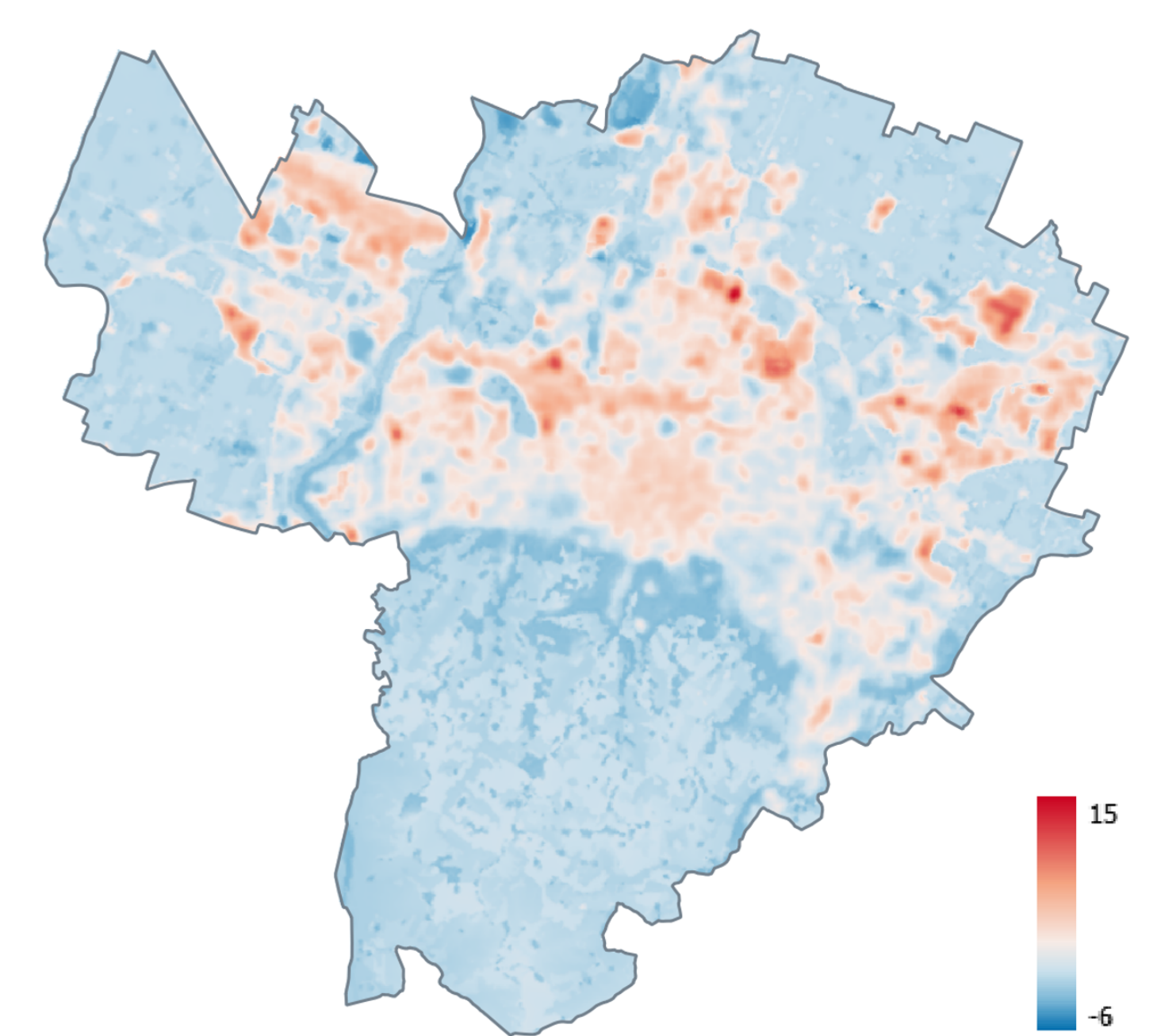


Results

- Posterior Mean Surfaces of the SV Intercept GRFs**

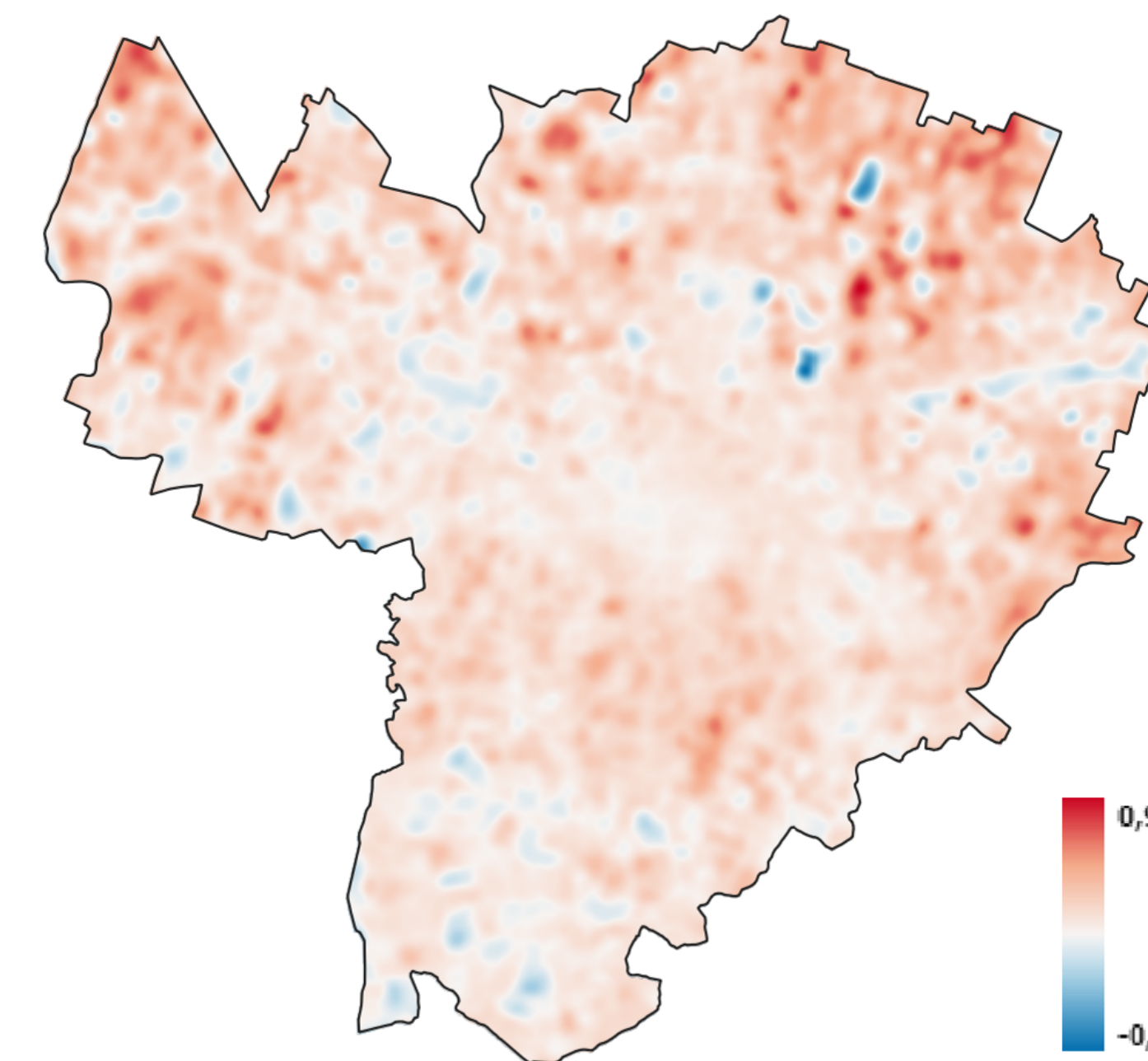


LST model

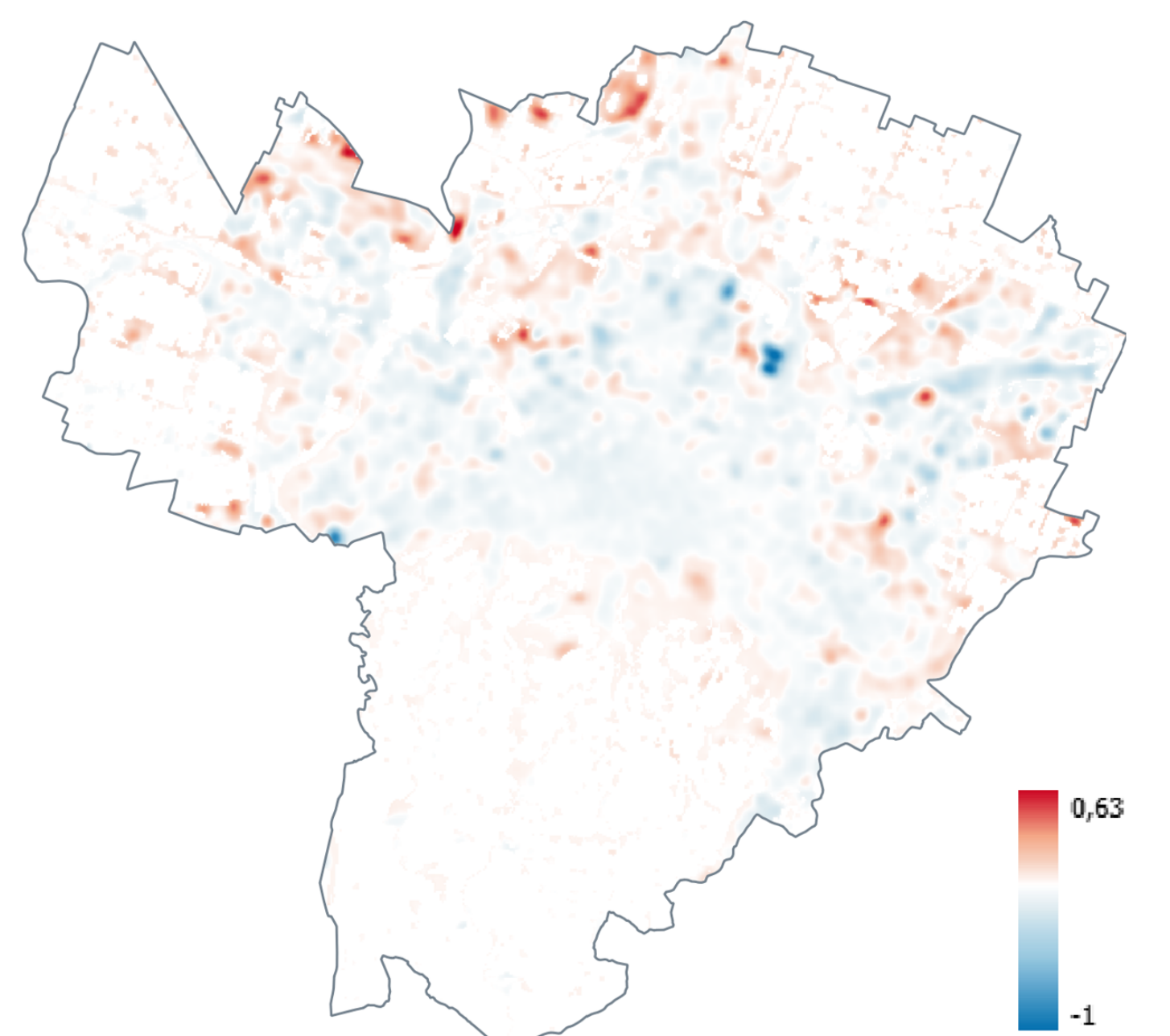


SUHI model

- Posterior Mean Surfaces of the SV Temporal Linear Trend GRFs**



LST model



SUHI model

- Posterior Excursion Sets at 95% Probability - Warming Regions**



Fig. 5: LST Model

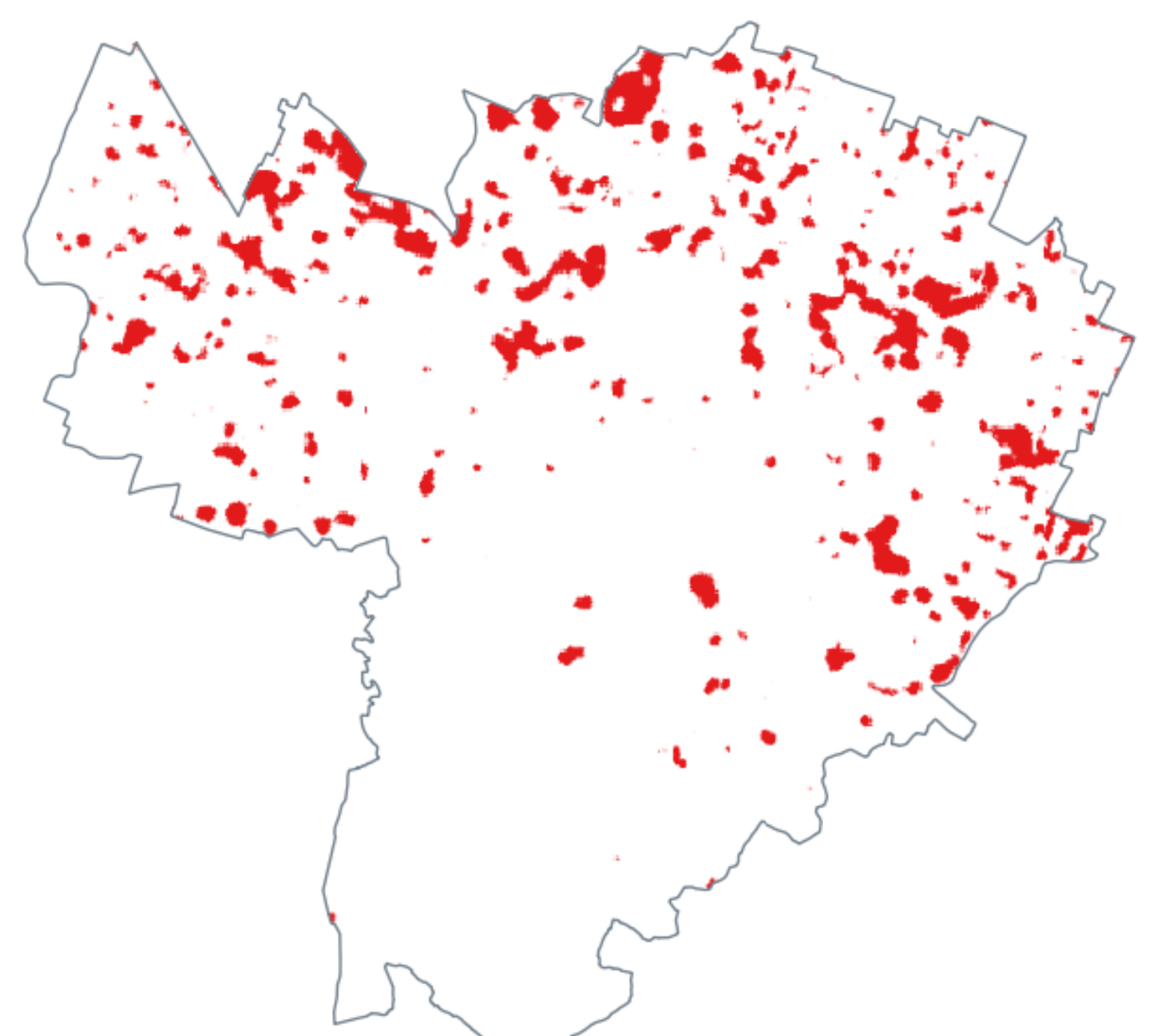


Fig. 6: SUHI Model

KEY REFERENCES

- [1] Bolin D., & Lindgren F. (2015) Excursion and Contour Uncertainty Regions for Latent Gaussian Models. *Journal of the Royal Statistical Society, Series B, Statistical Methodology*, 77(1):85-106.
- [2] Krainski E.T. et al. (2019) *Advanced Spatial Modeling with Stochastic Partial Differential Equations Using R and INLA*. Chapman & Hall/CRC Press.

ACKNOWLEDGEMENT AND PARTNERS



GRASPA
ENVIRONMENTAL STATISTICS

MORE INFORMATION



Marika D'Agostini

Department of Statistical Sciences
"Paolo Fortunati"

University of Bologna

marika.dagostini2@unibo.it